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REGULAR PAPER

An on-chip glitchy-clock generator for testing fault injection attacks

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Abstract This paper presents a glitchy-clock generator integrated in FPGA for evaluating fault injection attacks and their countermeasures on cryptographic modules. The proposed generator exploits clock management capabilities, which are common in modern FPGAs, to generate clock signal with temporal voltage spike. The shape and timing of the glitchy-clock cycle are configurable at run time. The proposed generator can be embedded in a single FPGA without any external instrument (e.g., a pulse generator and a variable power supply). Such integration enables reliable and reproducible fault injection experiments. In this paper, we examine the characteristics of the proposed generator through experiments on Side-channel Attack Standard Evaluation Board (SASEBO). The result shows that the timing of the glitches can be controlled at the step of about 0.17 ns. We also demonstrate its application to the safe-error attack against an RSA processor.

Keywords Fault injection attacks · Clock glitch · RSA · Safe-error attack

1 Introduction

Fault injection attacks are attracting much attention in the field of cryptographic hardware and embedded systems. The attackers first inject faults to cryptographic operations, and then estimate a secret key from the several faulty ciphertexts.

Boneh et al. [1] first proposed Bell-core attack against RSA cryptosystems based on Chinese Reminder Theorem (CRT). The attack calculates the difference between correct and faulty (i.e., fault-injected) outputs, which makes it possible to factorize the modulus used in RSA encryption/decryption. Yen and Joye then proposed Safe-error attack against public-key cryptosystems with Multiply-and-squaring-always method which is a conventional countermeasure to timing attack (TA) and simple power analysis (SPA) [2]. The attack is undetectable by common recalculation schemes performed just before the output since the key information is revealed when the fault is injected into a dummy operation.

After the first publication focusing on public-key cryptosystems [1], the fault injection attacks were also extended to symmetric-key cryptosystems. Differential fault analysis (DFA) [3] extracts secret key from the difference of correct and faulty outputs. In DFA, we assume that temporal faults are injected into some bits only for a short time and the faulty values are propagated through the following hardware logics without any additional faults. Ineffective fault analysis (IFA) [4] injects faults into specific operation and observe secret key from whether the output changes or not. Such advanced attacks sometimes require timely injected faults during specific operations.

With the advance of fault injection attacks and countermeasures, fault injection techniques have also been investigated to evaluate the possibility of the attacks in practice. Fault models are roughly categorized into two groups, namely permanent and transient faults. In a permanent fault, a target circuit is damaged forever. It is rather easy to detect such faults and react to them by means of power-on self-test (POST). On the other hand, a transient fault is induced at run time and the target circuit can be recovered at the end of

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